

HeatWave: Making MySQL Smarter

MySQL is a popular and powerful open source relational database management system. The inclusion of HeatWave in MySQL has enhanced its capabilities multifold. Read on to know more.



Oracle MySQL HeatWave (formerly called MySQL database service analytics) is an in-memory query accelerator developed at Oracle Labs, and provisions HeatWave clusters for accelerated processing of analytic queries. It is a very powerful service hosted on the cloud, eliminating the need for a separate analytics database and ETL processes.

Traditional analytics solutions: The challenges

Generally, customers have online transaction processing (OLTP) applications, which are connected to an OLTP database like MySQL. If users have to execute a query, they need to move their data to an online analytical processing (OLAP) database. This is done through extraction, transfer and load (ETL), and typically

requires a database administrator's (DBA) involvement. A DBA assists in migrating data from OLTP to OLAP, performs the query execution and sends the results back to OLTP. However, there are a few challenges existing solutions face:

- Running analytic queries against an OLTP database is always slow.
- The ETL processes with separate databases for OLTP and OLAP are lengthy.
- The data that is being processed is not up-to-date.
- The solutions are complex and expensive, as you need two databases and have to pay for two different services along with an experienced DBA.

Oracle MySQL HeatWave

MySQL HeatWave has emerged as a single solution to all the problems mentioned above. Both OLTP and OLAP are integrated adequately, and a built-in real-time query accelerator has been

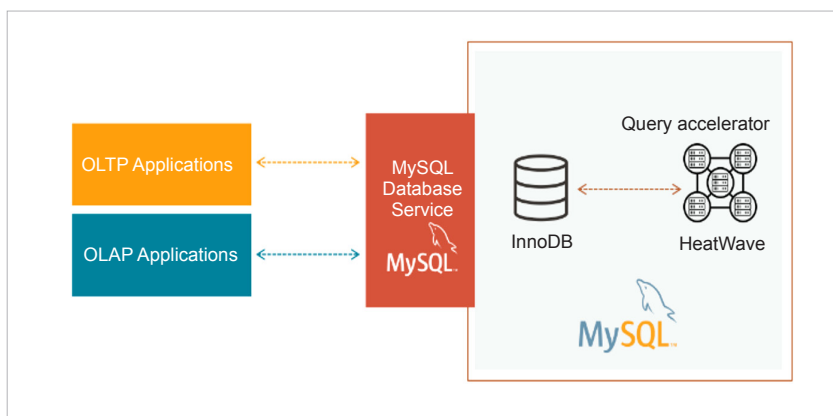


Figure 1: Architecture for HeatWave integration